

The empirical recurrence for OEIS A204613, proved by transfer matrix

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2 September 2026

Abstract

OEIS A204613 counts arrays over $\{0, \dots, 1\}$ under a condition on the determinant of every 2×2 subblock, and carries an empirical recurrence of order 23 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The condition is local to a bounded window of consecutive rows, so the count is a walk count on $S = 907$ states and is C -finite; whether the conjectured recurrence annihilates it is then settled by a finite exact computation, with the Cayley–Hamilton theorem bounding the work.

2020 Mathematics Subject Classification. 05A15, 05B45, 15B36.

1 The conjecture

OEIS A204613 is “Number of $(n+1) \times 6$ 0..1 arrays with the determinants of 2×2 subblocks nondecreasing rightwards and downwards.”. It has offset 1 and begins

907, 14199, 205117, 3084029, 45645661, 681254361, 10133992449, 150976448169, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: $a(n) = 36*a(n-1) - 406*a(n-2) + 404*a(n-3) + 23747*a(n-4) - 145816*a(n-5) - 129968*a(n-6) + 3736568*a(n-7) - 7902612*a(n-8) - 29701104*a(n-9) + 128249776*a(n-10) + 34934976*a(n-11) - 780114560*a(n-12) + 607369728*a(n-13) + 2210927104*a(n-14) - 3052507136*a(n-15) - 2863537152*a(n-16) + 5709414400*a(n-17) + 1514283008*a(n-18) - 4781309952*a(n-19) - 281149440*a(n-20) + 1658847232*a(n-21) + 94371840*a(n-22) - 150994944*a(n-23)$ for $n > 26$.

As of the “Last modified” line on the live entry (Jan 01 2024 19:47:38, revision 7) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

Write σ for the determinant of a 2×2 subblock $\begin{pmatrix} p & q \\ r & s \end{pmatrix}$, that is $\sigma = ps - qr$. The subblocks of an $(n+1) \times 6$ array over $\{0, \dots, 1\}$ form a grid of n rows and 5 columns, and σ colours that grid.

The condition relates NEIGHBOURING subblocks: the values of two subblocks adjacent in the grid, horizontally or vertically, must not decrease. Rightwards and downwards are the two directions named.

3 The count is a walk count

A subblock's value needs two consecutive array rows, and the vertical comparison needs the two subblock rows that three consecutive array rows produce. So the state is the PAIR of consecutive array rows: the horizontal comparisons decide which pairs are admissible at all, and the vertical ones decide which pair may follow which. A step appends one array row, turning (r, s) into (s, t) and settling every comparison between the two subblock rows at once. An $(n + 1)$ -row array is a walk of $n - 1$ steps, so

$$a(n) = \iota^\top M^{n-1} \tau, \quad \iota = \tau = (1, \dots, 1)^\top,$$

on the $S = 907$ admissible pairs.

At $n = 1$ the array has two rows, so the grid is a single row of 5 subblocks with no vertical neighbours, so only the horizontal comparisons apply. The entry's first term is 907, and the model reproduces it along with every later term; that is the cheapest check that the grid has been read with the right shape.

Over the alphabet $\{0, \dots, 1\}$ the determinant of a 2×2 subblock takes values in $[-1, 1]$, so the colouring of the grid it induces has at most 3 values. That bound is what keeps the construction below finite in the only place where the value of σ , rather than a comparison between two of them, has to be carried. In particular a is C -finite. The alignment of the walk index with the entry's own n was checked against its published terms rather than assumed.

4 The proof

Lemma 1. *Let $q(t) = t^{23} - \sum_i c_i t^{23-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j , and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+23} - \sum_i c_i M^{j+23-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M ; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 2.

$a(n) = 36a(n-1) - 406a(n-2) + 404a(n-3) + 23747a(n-4) - 145816a(n-5) - 129968a(n-6) + 3736568a(n-7)$
for every $n > 26$.

Proof. The vector $w = q(M)\tau$ was formed by 23 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots$, again exactly; those integers vanish from the index corresponding to $n = 26 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. $\square$

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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- [3] H. Minc, *Permanents*, Addison–Wesley, 1978.
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.